

Fidelity Witness

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Last Update: September 20, 2026

1 Findings

- we rely on the inequality

$$\forall O : F(\rho_t, \rho_p) \geq F(\rho_t, O)^2 + F(O, \rho_p)^2 - 1$$

- Using [classical shadows framework](#), by collecting multiple measurements from ρ_t and ρ_p we can have an estimate of $F(O, \rho)$ for any O as long as O has a simple structure like being a pure separable state.

A classical gradient descend can find us the O that produces a relatively high lower bound for the fidelity.